

Motivation

- Students with low financial literacy face significant barriers to accessing relatable and engaging education, which hinders their ability to develop sustainable management skills.
- Limited literacy can perpetuate economic inequality and reduce a student’s confidence in planning for both short-term and long-term goals.
- Static, one-size-fits-all curricula often fail to address individual learning needs or adapt to a learner’s lived experiences.
- Integrating gamification and explainable AI (XAI) creates a reflective environment where students can experiment with financial trade-offs without real-world risk.

Theoretical Background

- Narrative simulations support learning by replacing abstract financial puzzles with concrete cause-and-effect scenarios for risk-free experiential reflection [1,2].
- XAI research shows that narrative storytelling can reduce cognitive load and enhance trust, while factual framing is effective for showing hidden system complexity [3,4].

Research Objectives

- RQ1: How does Narrative vs. Factual AI explanation framing impact comprehension and application of financial principles among college-age students?
- Analytical vs. Storytelling: This study treats AI as both an analytical tutor (Factual) and a storytelling partner (Narrative) to determine which better promotes reflection and motivates sustained engagement.

Methodology

- Design: Quasi-experimental, within-subjects study ($N = 20$) involving three distinct phases of data collection.
- Phases:
 - Pre-intervention: Establishing a baseline of basic financial literacy.
 - Mid-intervention: Assessing comprehension of common financial concepts within the levels.
 - Post-intervention: Measuring information retention and conducting a verbal debrief.
- Participants: All participants were recruited from Wellesley College, a 100% need-met institution where 56% of students are on financial aid.

The “What-If” Simulation

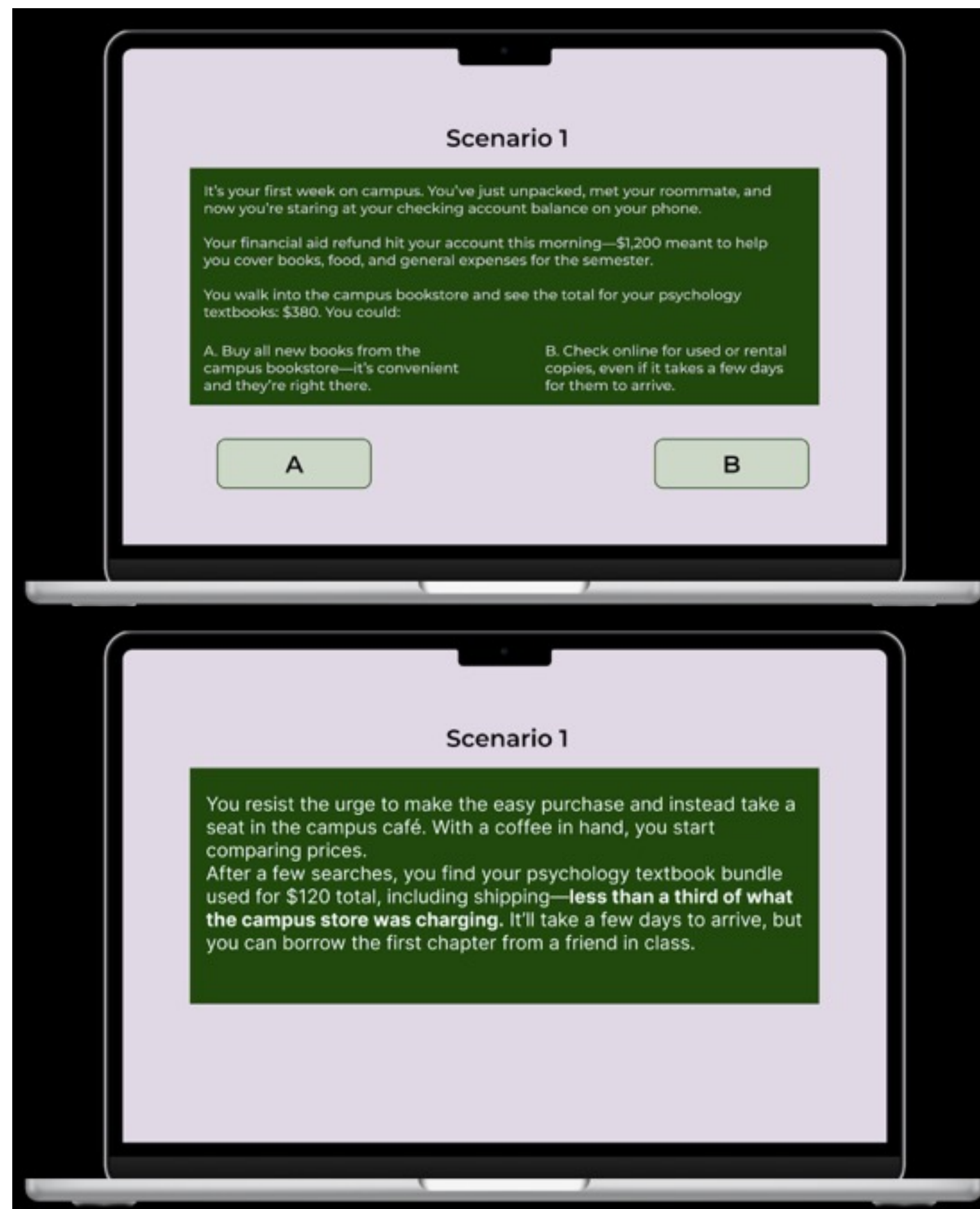


Figure 1. Figma prototypes of one scenario and its explanation in the Narrative mode on a desktop view.

- Interactive System: A desktop-based game where players navigate real-world financial scenarios to explore long-term consequences in a risk-free environment.
- AI engine: Two customized GPT models generated the feedback, framed in either a Narrative or Factual condition.

XAI Framing Conditions

Narrative Condition	Factual Condition
- Story-like reflections on the player’s financial journey. - Focuses on emotional and practical implications.	- Concise, data-driven feedback (balances, interest rates). - Focuses on objective cause-and-effect metrics.

References

- [1] Santos et al. (2025). *DreamScape*: Fostering Financial Cognition. C&C '25.
[2] Weichselbraun et al. (2025). Cash Action: Educational Multiplayer Game for Financial Concepts. FDG '25.
[3] Martens et al. (2025). Tell Me a Story! Narrative-Driven XAI with LLMs. *Decision Support Systems*.
[4] Celar & Byrne (2023). Reasoning with Counterfactual and Causal Explanations for AI. *Memory & Cognition*.

Quantitative Results

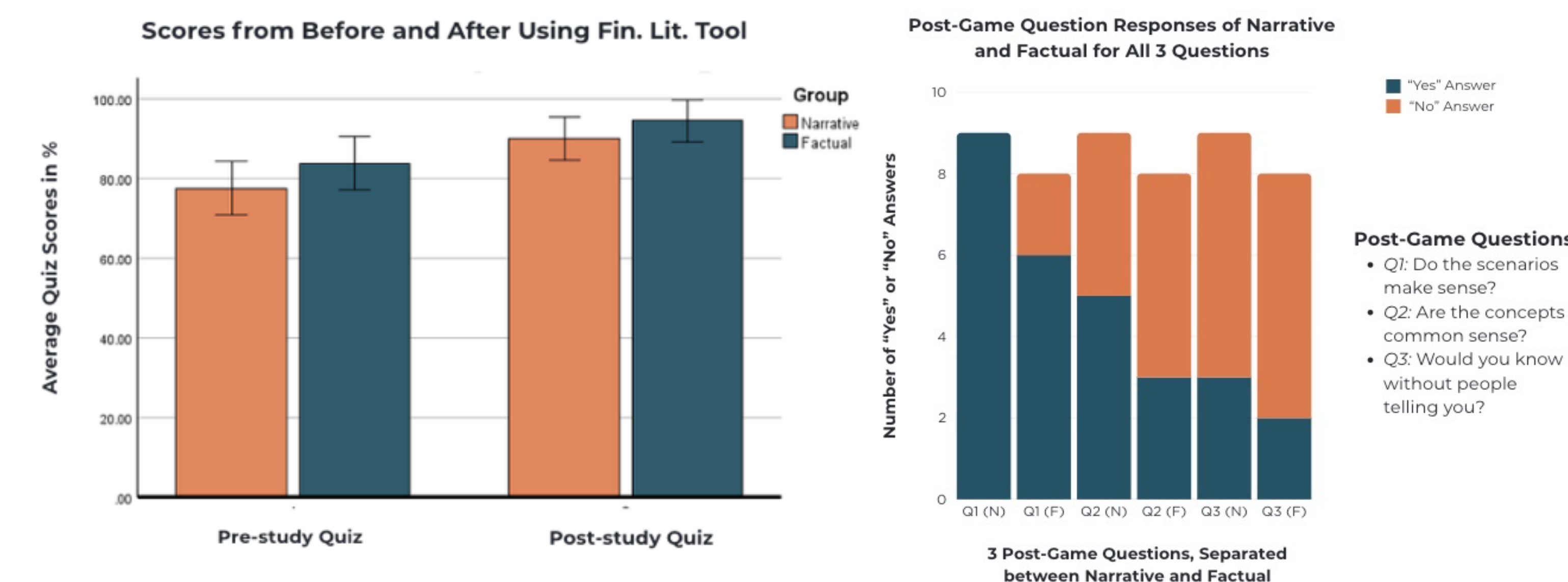
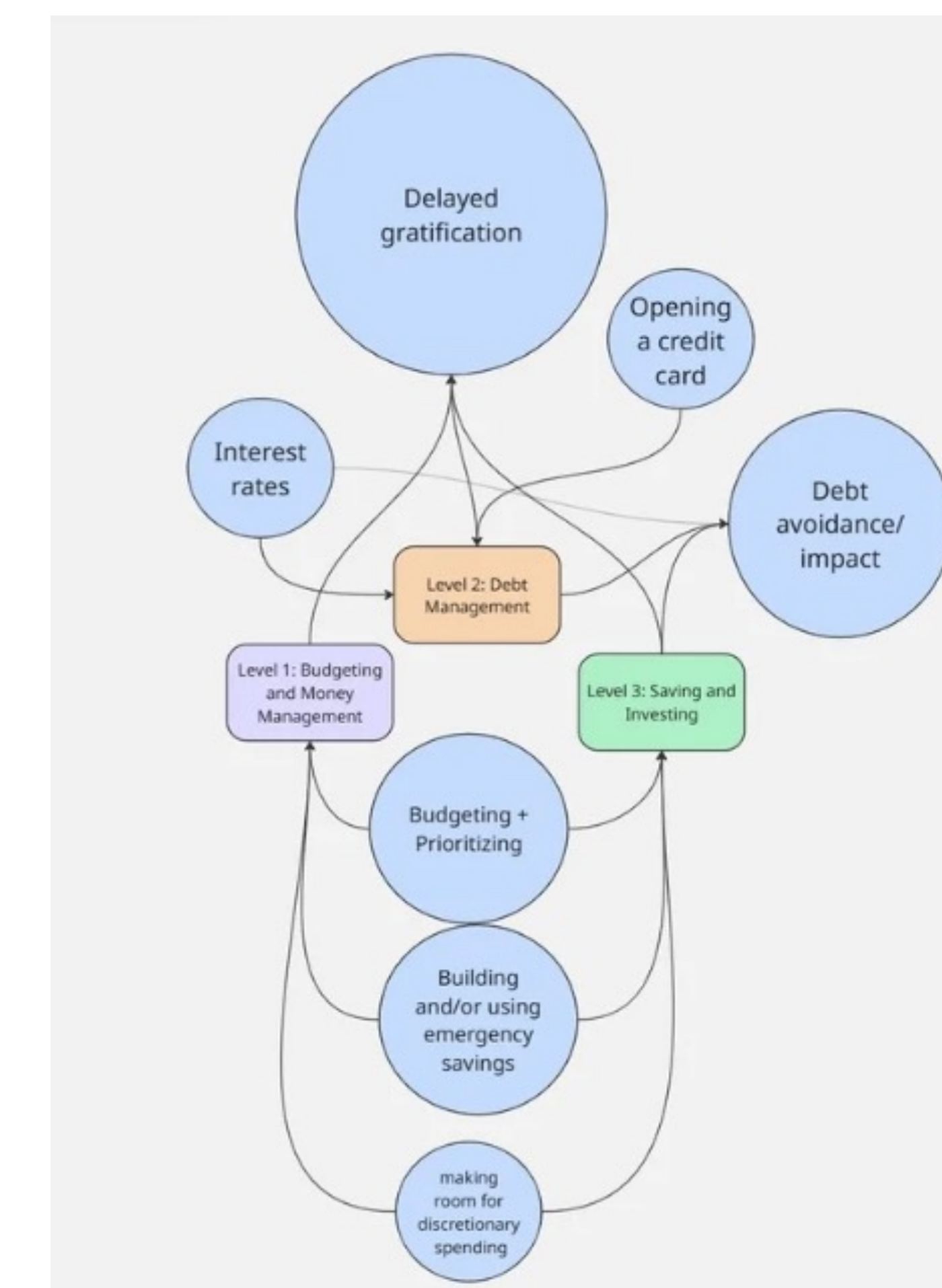


Figure 2 (left). Bar chart showing no statistically significant improvements in financial knowledge between baseline and retention surveys for both Narrative and Factual conditions

Figure 3 (right). Stacked bar chart showing the number of “yes” and “no” responses during the post-study verbal debrief for both Narrative and Factual conditions

- No statistically significant improvements in objective financial knowledge gains between conditions.
- Factual condition participants were less likely to label the financial concepts as “common sense”.
- Factual XAI may better highlight hidden financial complexity.

Qualitative Results



- Delayed Gratification: The core takeaway across all three levels.
- Debt management (Level 2) provided the most specific technical learning, with students citing new knowledge (interest rates, credit card mechanics).
- Participants highlighted the importance of building emergency funds while balancing discretionary spending.
- Some users noted that the absence of “clear right or wrong” answers in certain scenarios made identifying a singular takeaway challenging.

Design Implications and Future Work

- Explanation styles shape how learners interpret knowledge, even if short-term scores don’t change.
- Future tools should focus on real-world predicaments (credit, interest), where students are most receptive to technical help.

Acknowledgements

We would like to thank our professor, Dr. Orit Shaer, for her guidance and feedback throughout the development of this work. We also thank our teaching fellow, Becky Chen, for her continued support and assistance.